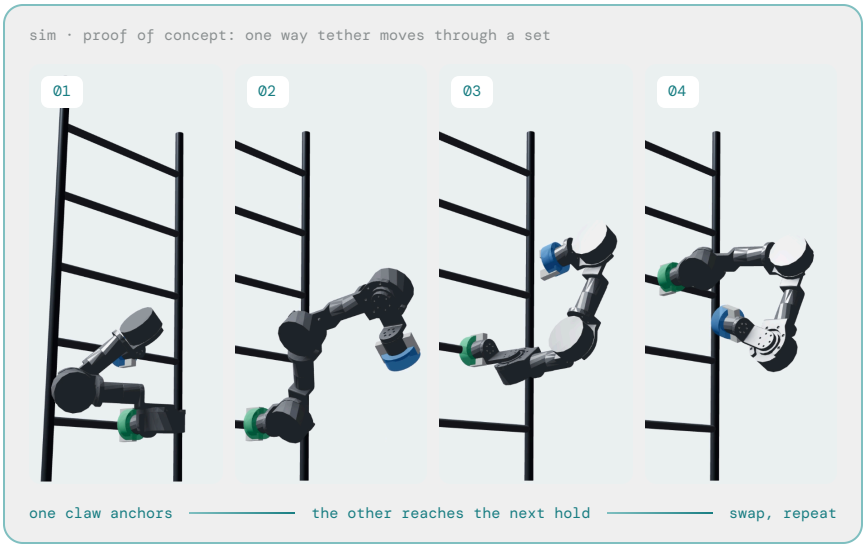
tether.

01 vision.



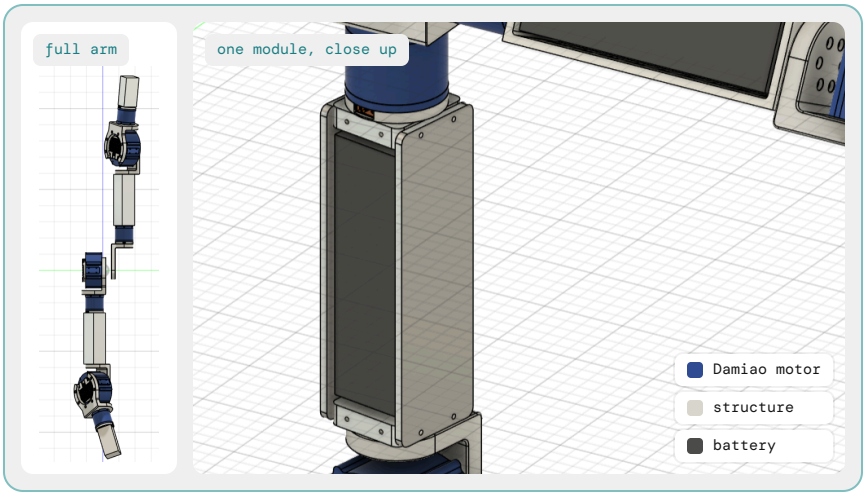
Tether is a redesign of the 7-DOF robot arm we built last semester. It becomes an inchworm robot with a claw on both ends.

One end grips while the other swings to the next hold, then they swap. It needs no fixed base and no cable to a wall.

On Demo Day, Tether moves through the demo set: climbing stairs, crossing between holds, or whatever the set calls for. The sim's ladder climb is a proof of concept of how it moves, not the final set.

02 technical.

hardware

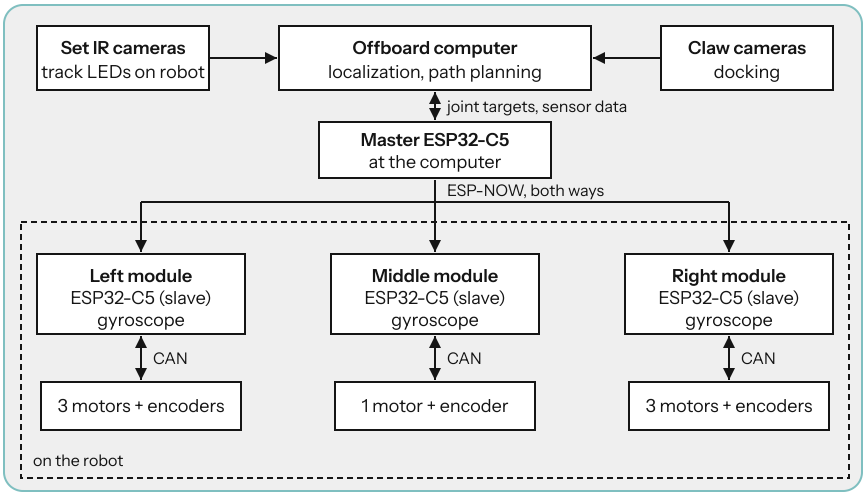


Move with no fixed base → one chain of seven Damiao motors (blue) with a claw on each end. One claw holds the robot in place while the other moves.

Joints that spin freely → no wires cross any joint. Wires or slip rings would limit rotation, so the robot is fully wireless and no cables show. The middle joint spins 360°.

No power cables through joints → three modules (left, middle, right). Each has its own 6S LiPo (dark grey) inside its structural link (light grey) and its own ESP32-C5.

software



Find itself in the dark → IR cameras fixed around the set track small IR LEDs hidden on each module. The lab's mocap can't move; tags fail in the dark and look bad.

Know its shape, grip precisely → encoders and a gyroscope on every segment fill in between camera frames; a camera on each claw lines up each grip.

Keep the robot small → all compute is offboard. Stage 1 runs pre-planned paths; RL and VLAs may follow.

Run three modules as one arm → a master ESP32-C5 at the computer talks to each module's ESP32-C5 over ESP-NOW. Each module drives its motors over CAN.